

		collision, the object will swap their velocities before and after collision
7	C	$\text{Weight} = (\text{Tension})\cos(30^\circ)$ $F = (\text{Tension})\sin(30^\circ)$ Hence, $F/\text{Weight} = \tan(30^\circ)$ $F = 170 \text{ N}$
8	B	By conservation of energy